

Double-Flip Sequential CFET: Superior 3T Library Efficiency and PPA

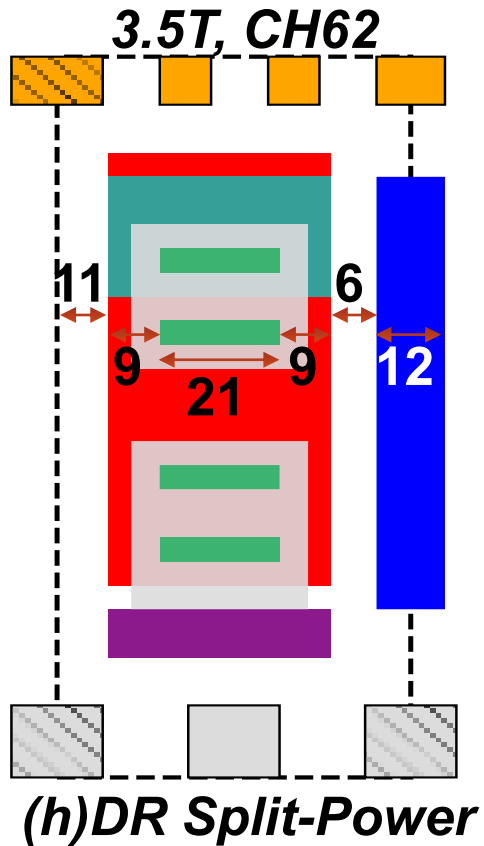
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imec, *equally contributed

Outline

- Introduction
- Double-Flip Sequential processing flow
- Standard Cell Library Layout Efficiency
- Power Performance evaluation
- Conclusions

Introduction



Monolithic



Sequential

Common gate



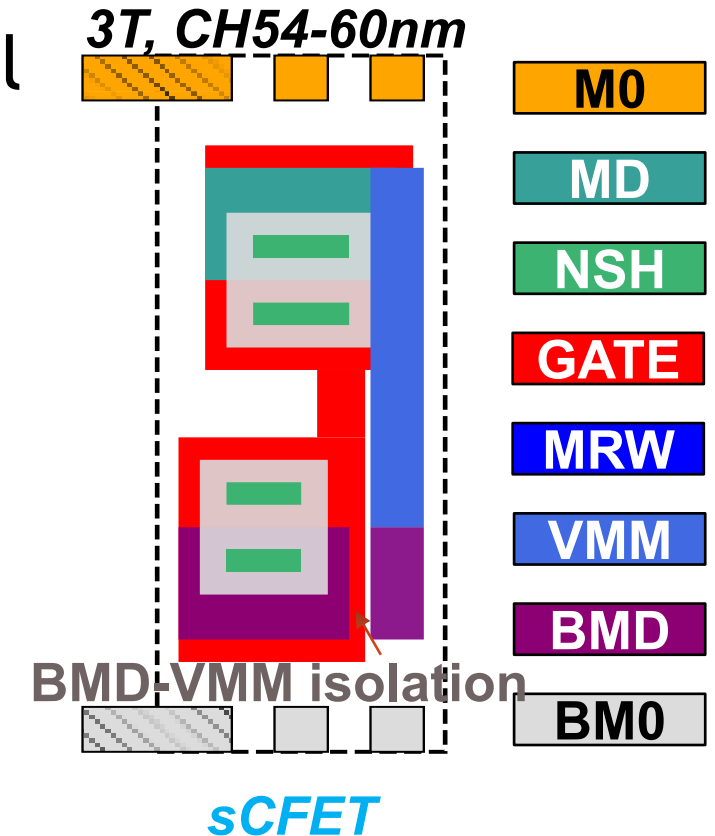
Split gate

3.5T

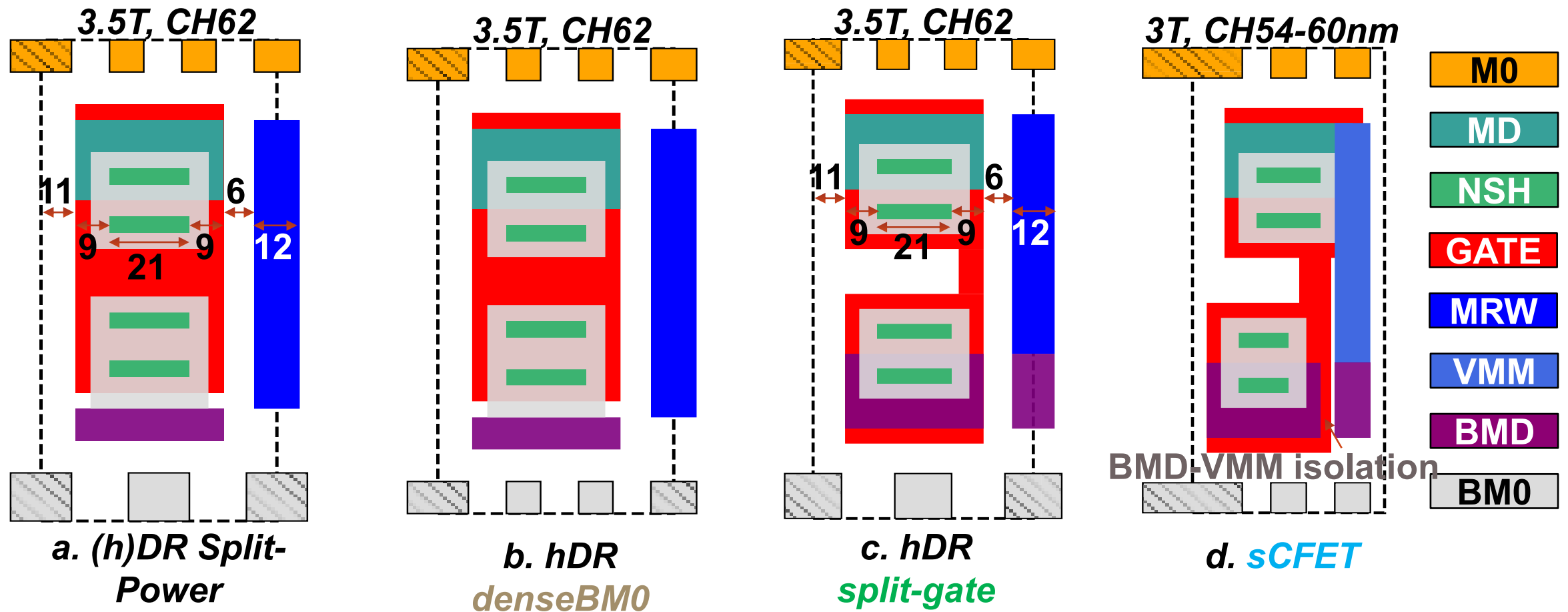


3T

- Processing
- Standard cell library
- Power performance



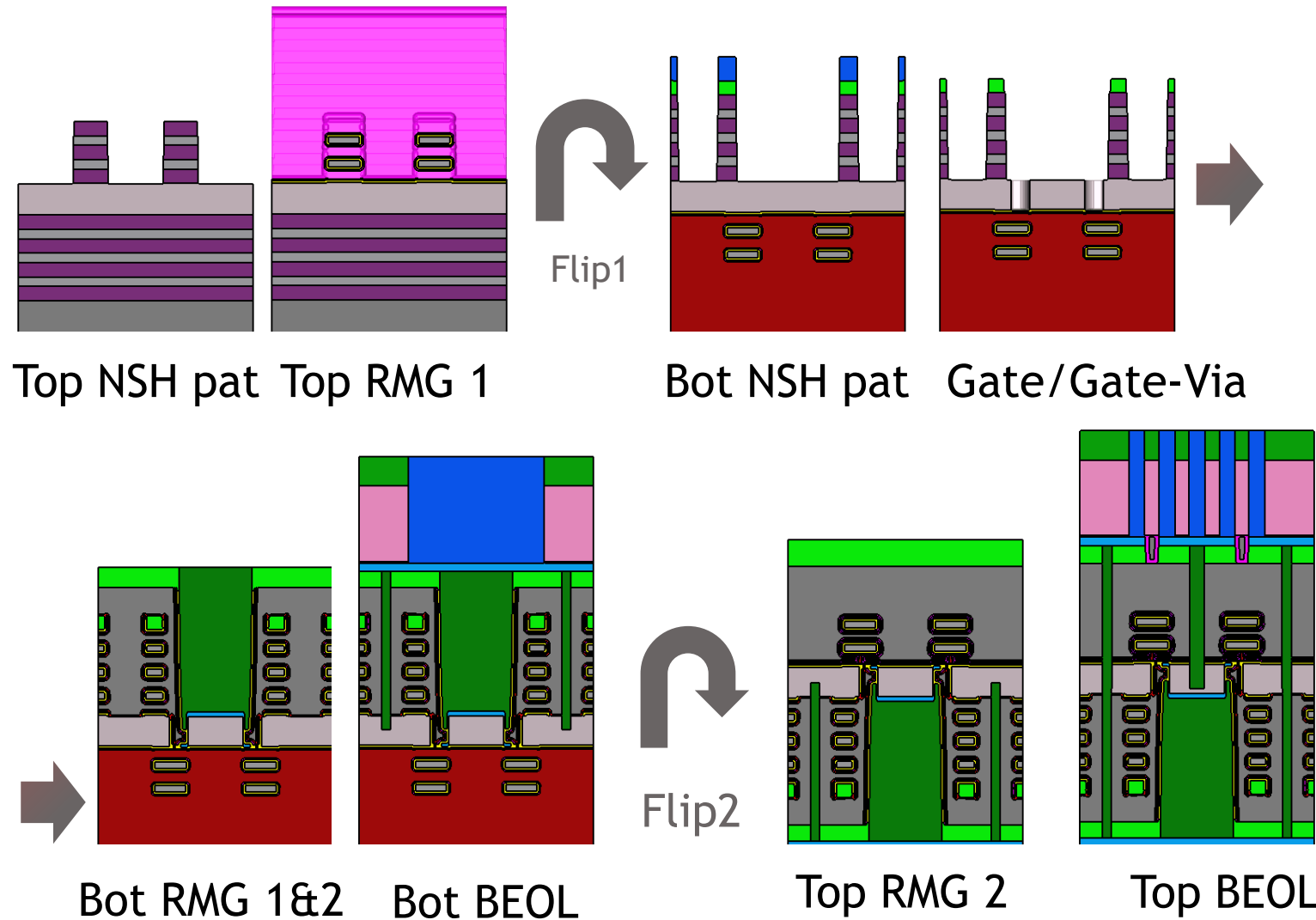
Introduction



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- Introduction
- **Double-Flip Sequential processing flow**
- Standard Cell Library Layout Efficiency
- Power Performance evaluation
- Conclusions

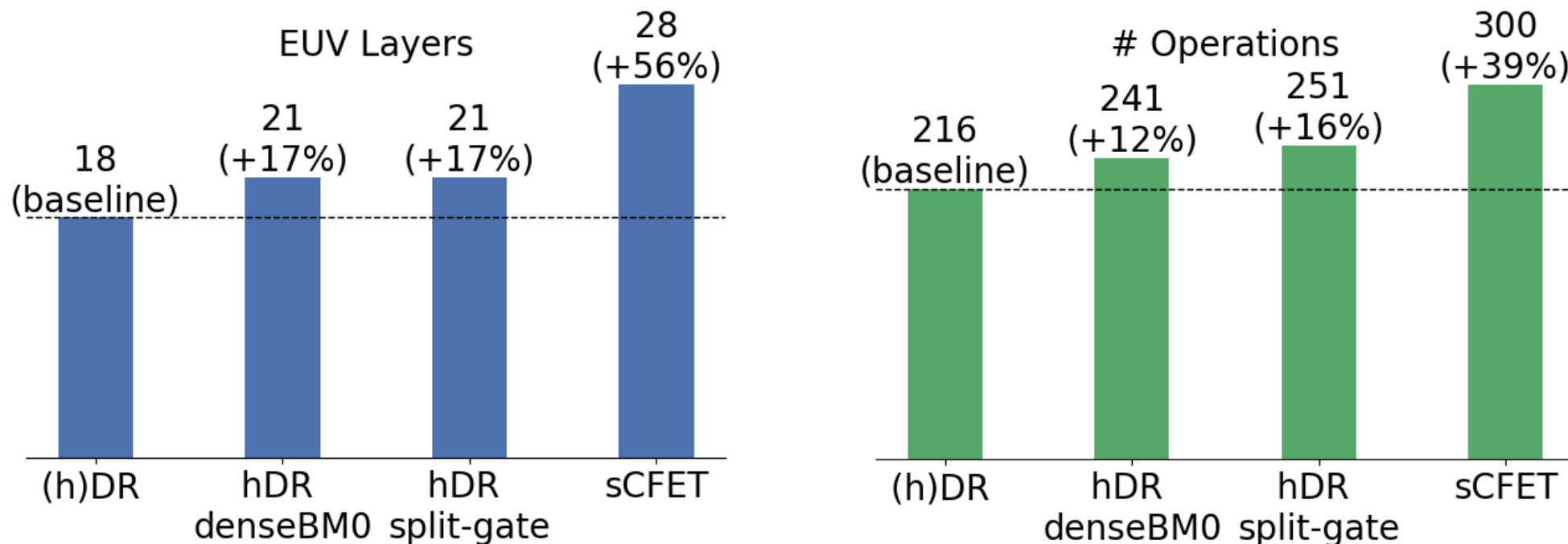
Double-Flip Sequential Processing flow



- Sequential patterning:
 - Reduce etch complexity
 - Avoiding High-aspect-ratio
- Two-step RMG:
 - Metal fill at low-T
 - Low-Vt for both tier

Flow comparison

	<i>(h)DR</i>	<i>hDR</i> <i>denseBM0</i>	<i>hDR</i> <i>split-gate</i>	<i>sCFET</i>
Integration	Monolithic	Monolithic	Monolithic	Sequential
BM0 pitch (nm)	31	18	31	18
Hybrid orient.	No	No	No	Yes
Split gate	No	No	Yes	Yes



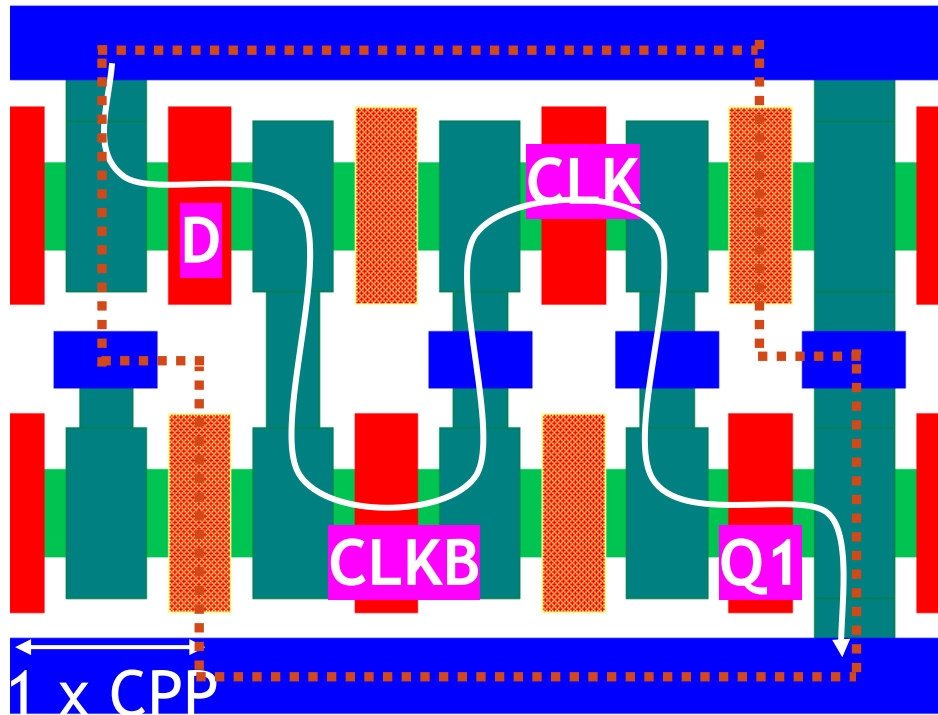
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Common gate vs Split gate

DFFD1

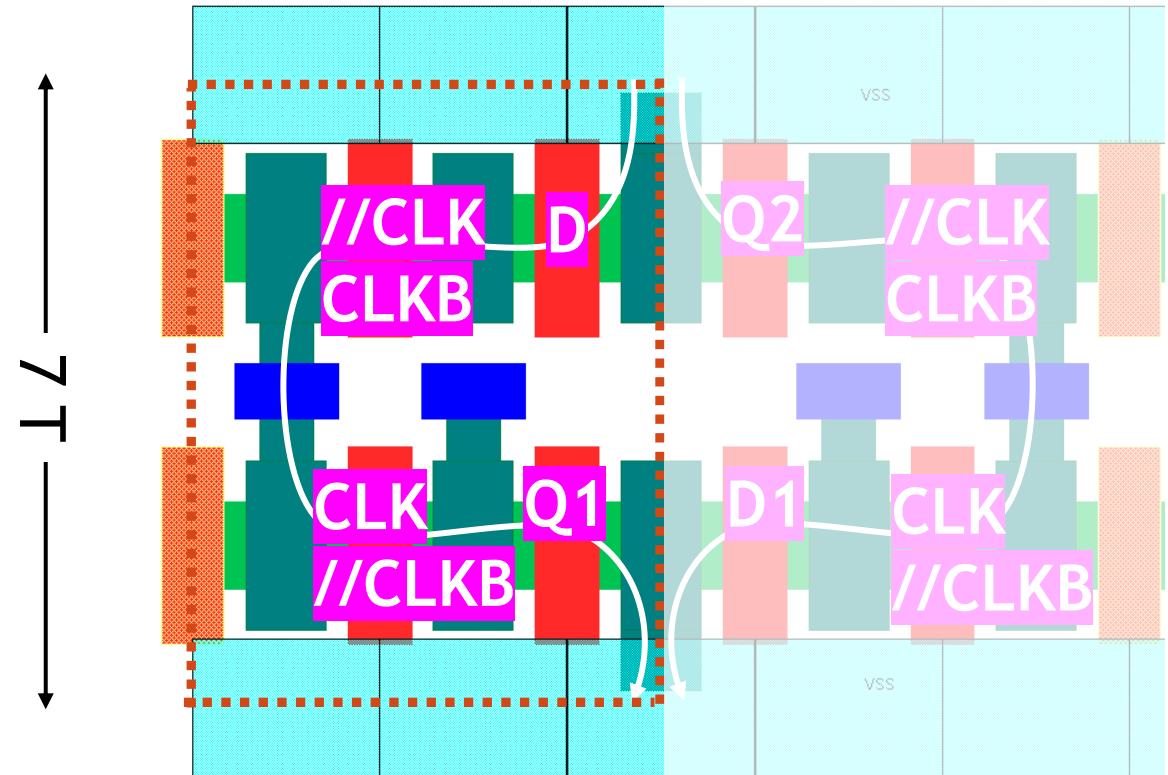
Common-gate (only master is shown)



1 master occupies $7 \times \text{CPP}$ — $-2 \times \text{CPP}$ —

GATE	DUMMY	MD	M0
NSH	MRW	BMD	M1

Split-gate (both master and slave are shown)

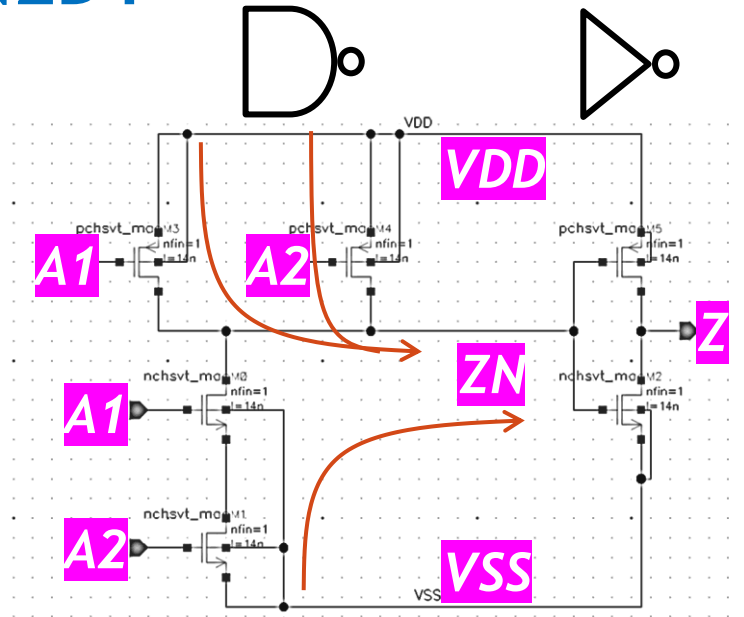


1 master occupies $5 \times \text{CPP}$

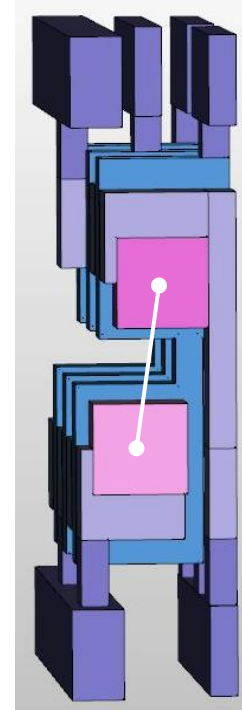
//: backside

Top MD to diagonal Bottom BMD connection

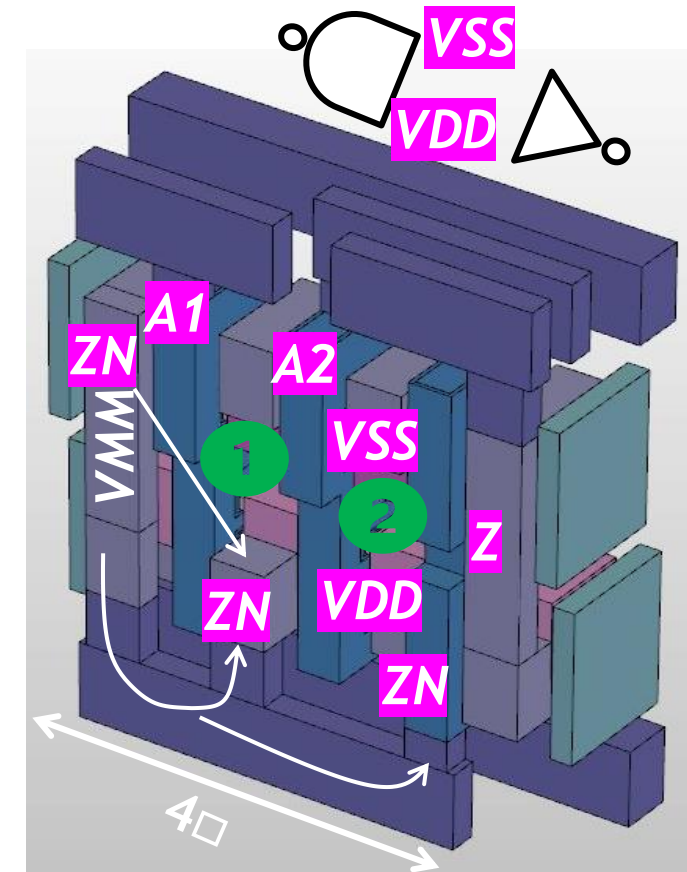
AN2D1



$$3\Box + 2\Box \neq 5\Box \\ = 4\Box$$



Offsetted
devices



Asymmetric VMM facilitates top MD $\nrightarrow$ bottom BMD diagonal connections

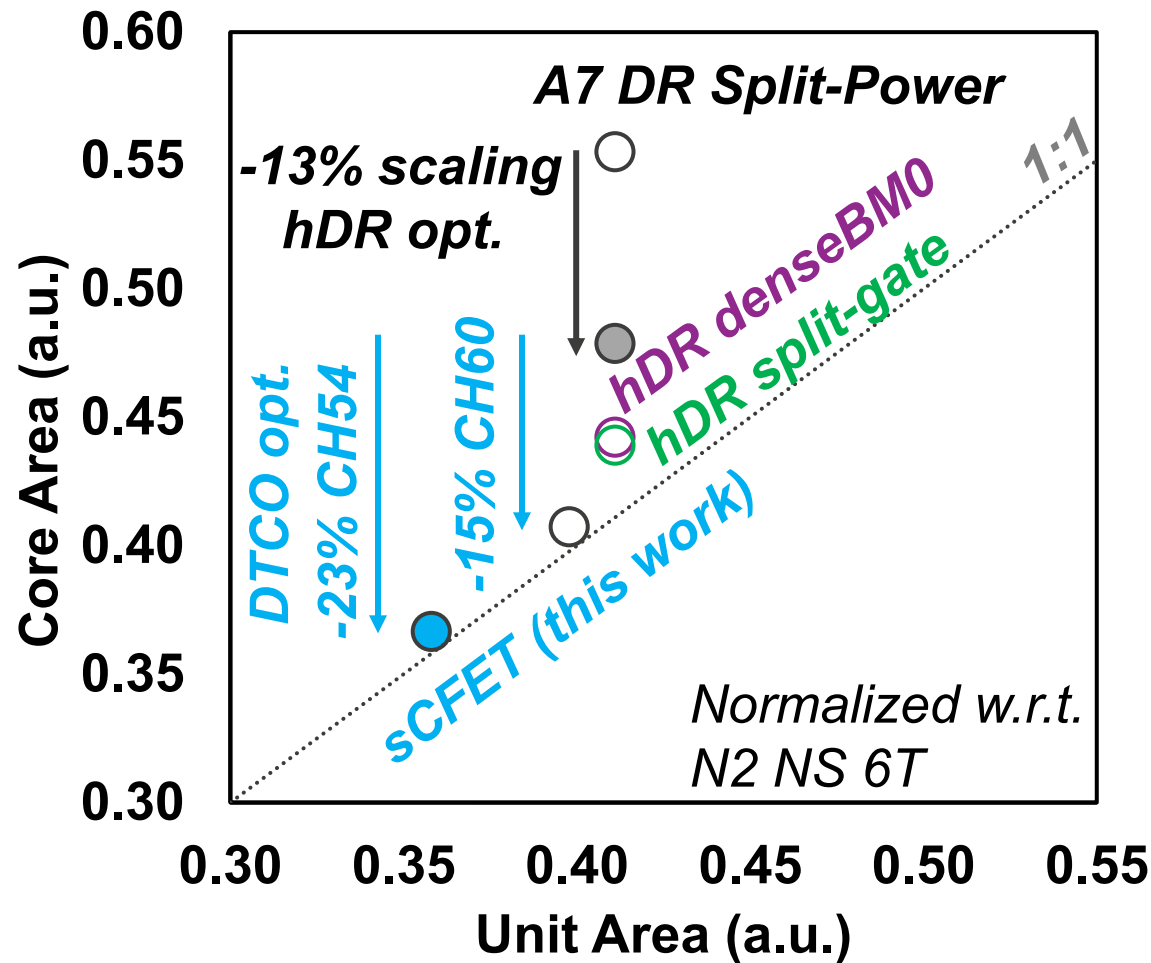
Standard Cell Library Layout Efficiency

split				common			Gate
18	31	18	31	31	BM0P (nm)		
54	62	62	62	62	CH (nm)		
sCFET This work	Split gate	hDR denseBM0	hDR	DR	CFET Arch		
					AN2D1		
					AN2D2		
					AO22D1		
					AOI22D1		
					BUFFD1		
					BUFFD4		
					DFFD1		
					DFFRLD1		
					DFFSLD1		
					FASTKD1		
					ICGD4		
					INVD1		
					INVD4		
					MUX2D1		
					ND2D1		
					ND2D2		
NR2D1							
NR2D2							
OA22D1							
OAI22D1							
OR2D1							
OR2D2							
XNR2D1							
XOR2D1							
0.98	0.94	0.93	0.86	0.74	η_{layout}		

Normalized w.r.t. N2 NS 6T

100% Δ area (rows × cols) 0%

Standard Cell Library Layout Efficiency



mCFET Double-Row CFET (*DR*)

→ significant area overhead

mCFET half-height DR CFET (*hDR*)

→ single-row like cell layout style

sCFET achieves

→ combined split-gate + denseBM0

→ highest layout efficiency (98%)

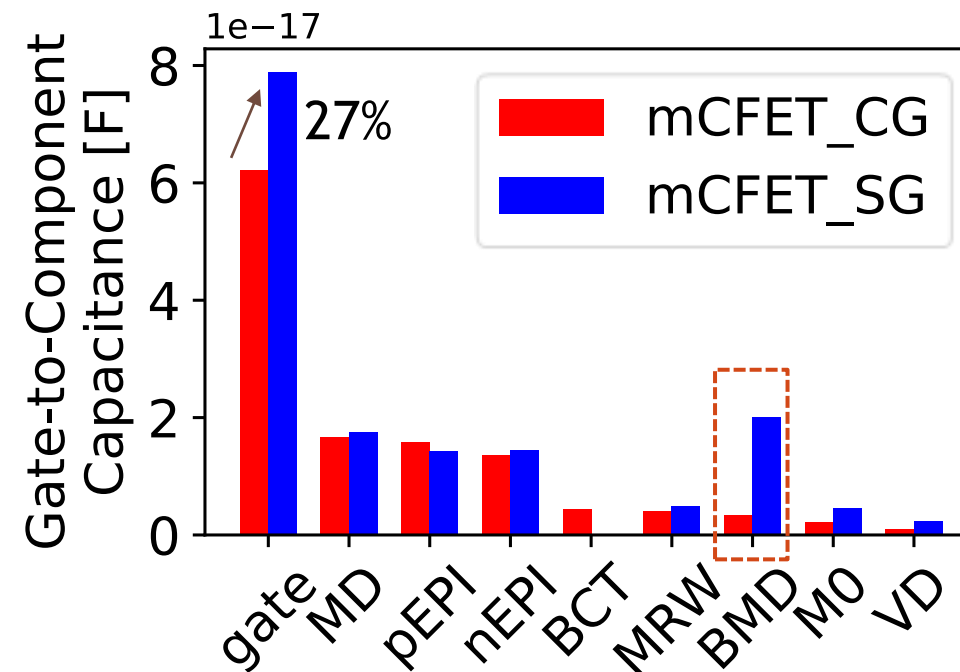
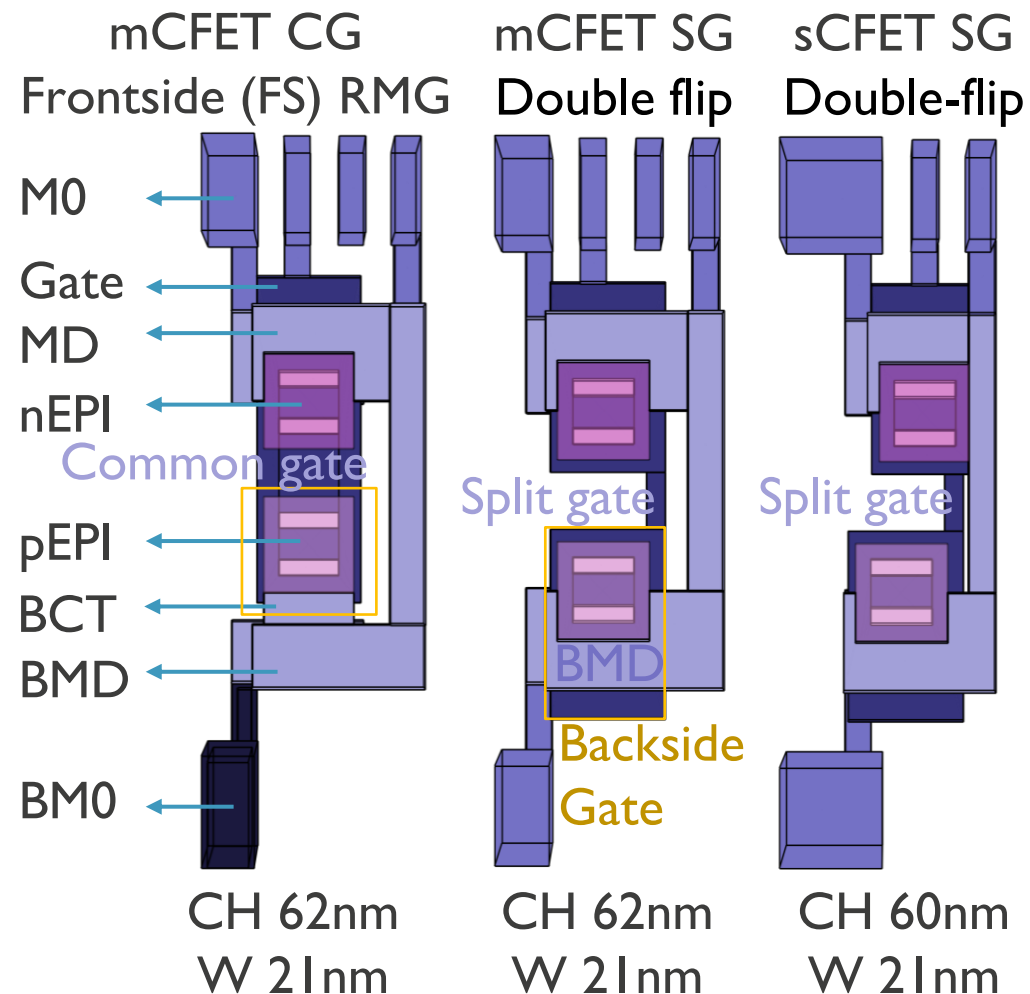
→ lowest core area (-23%)

$$\eta_{\text{layout}} = \frac{\text{scaling}_{\text{Core Area}}}{\text{scaling}_{\text{Unit Area}}}$$

Outline

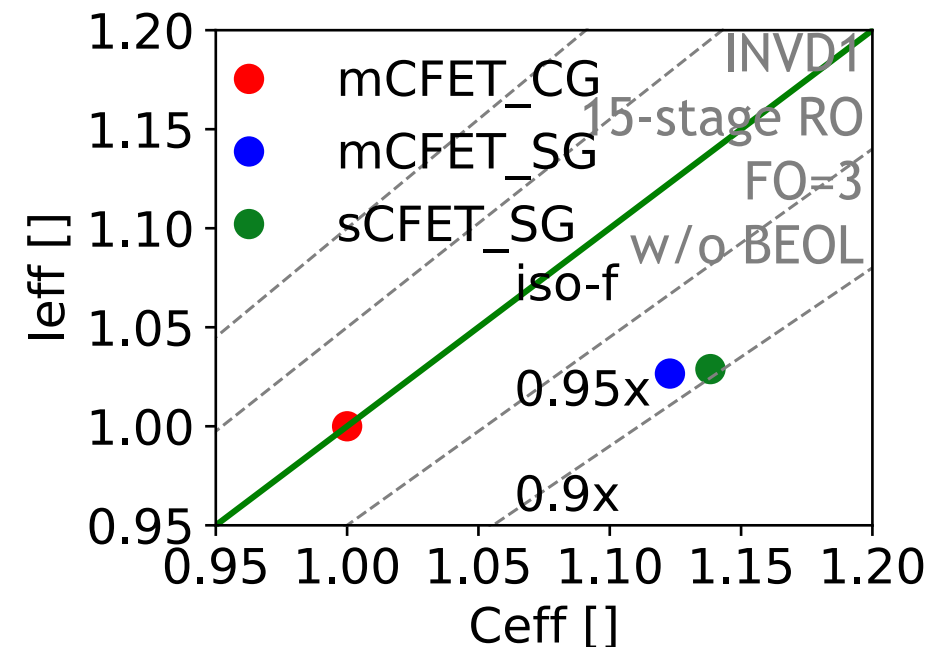
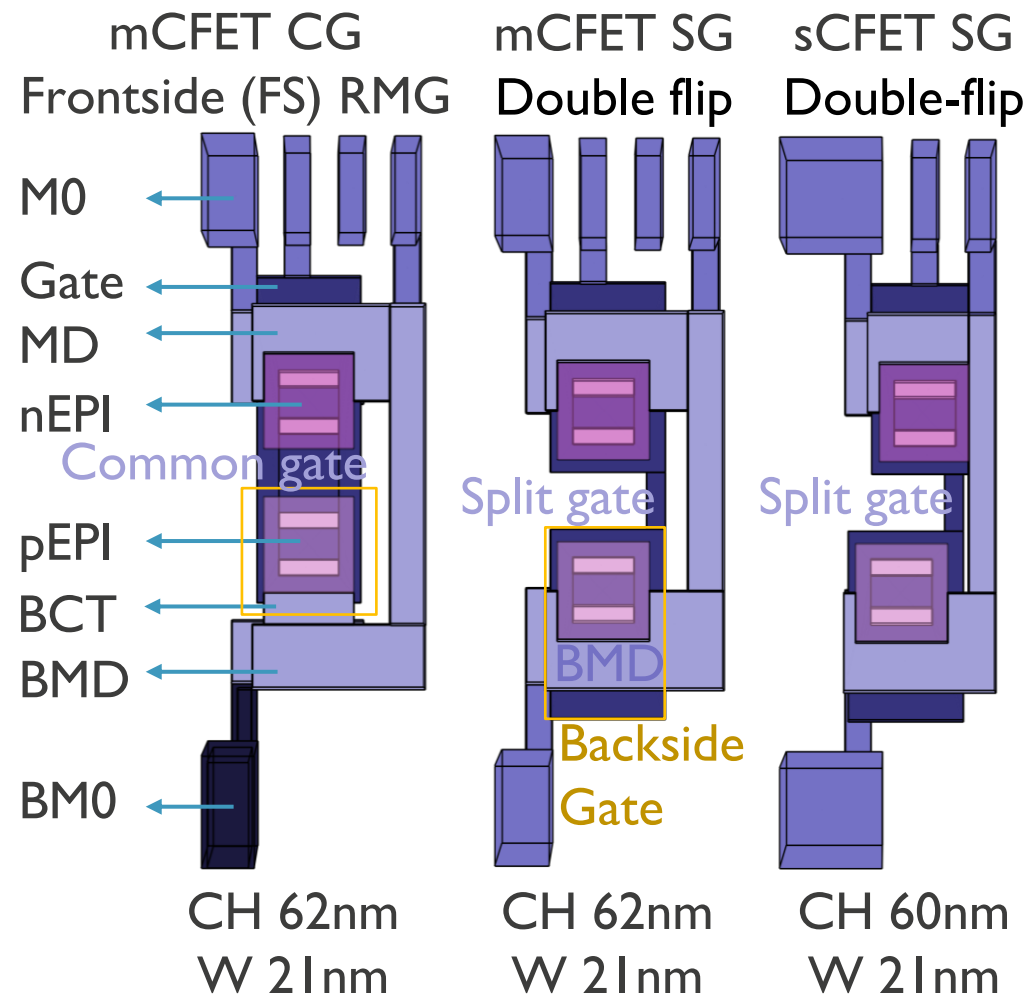
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Gate capacitance evaluation



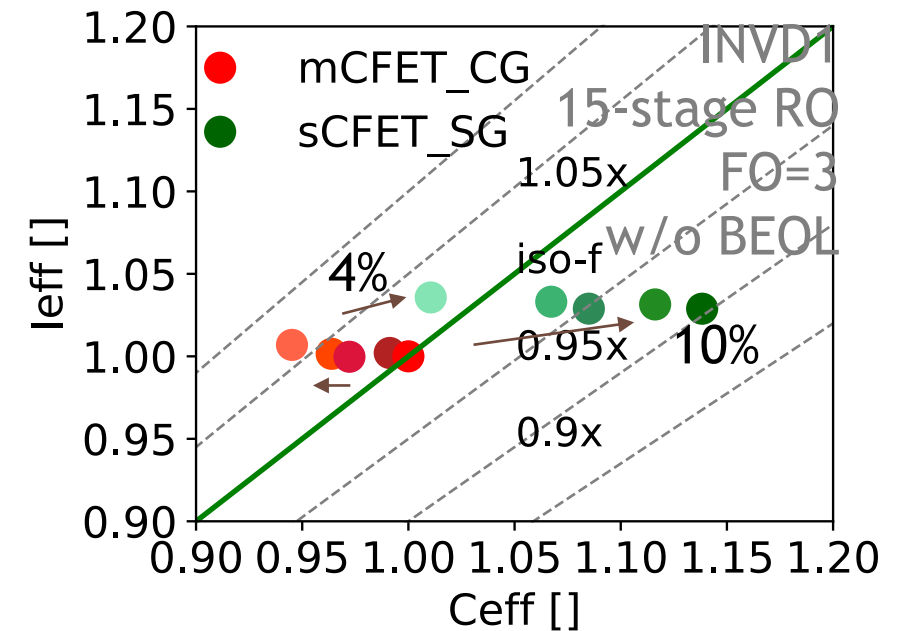
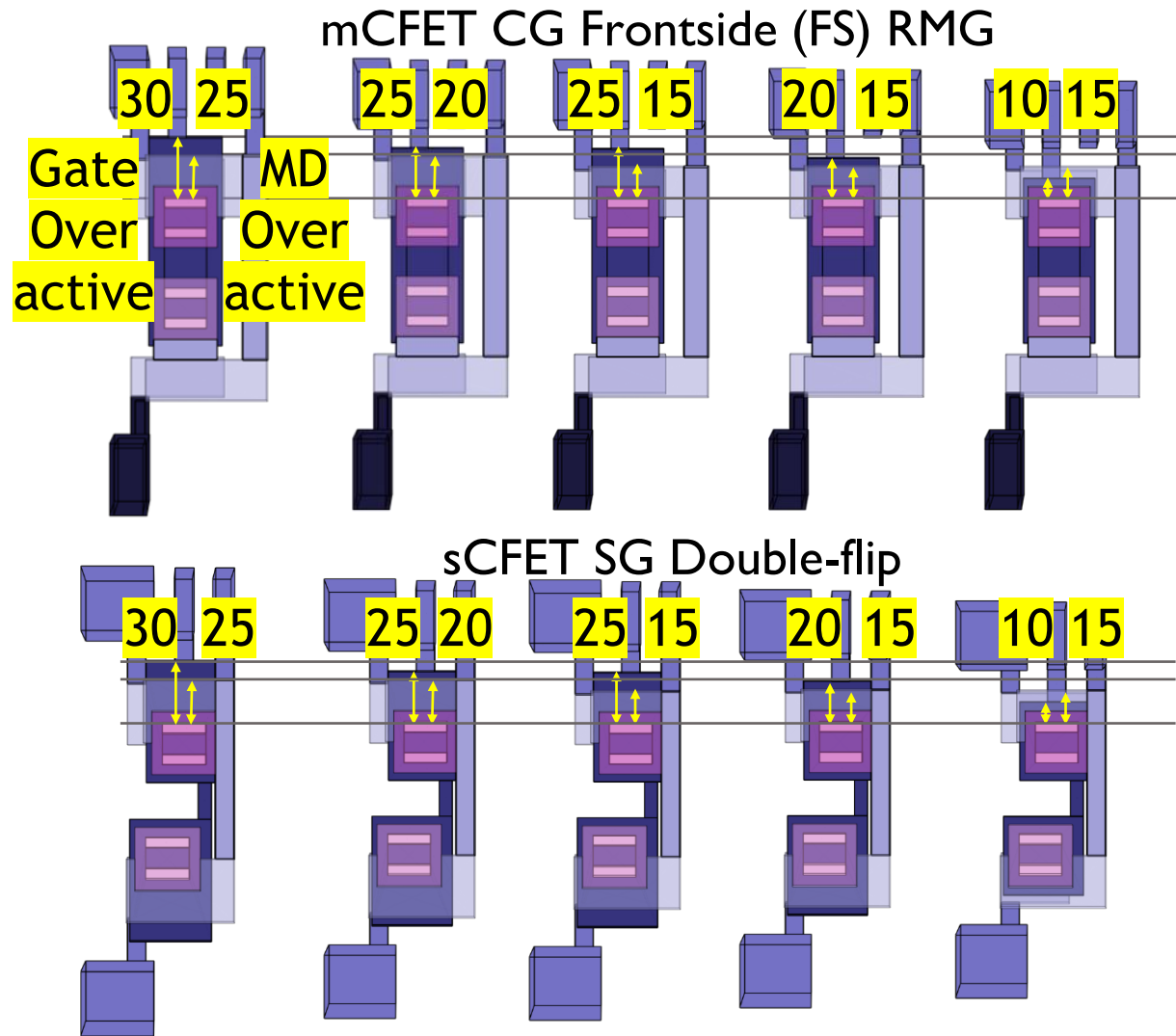
Taller backside gate increases parasitic gate capacitance

Gate capacitance evaluation



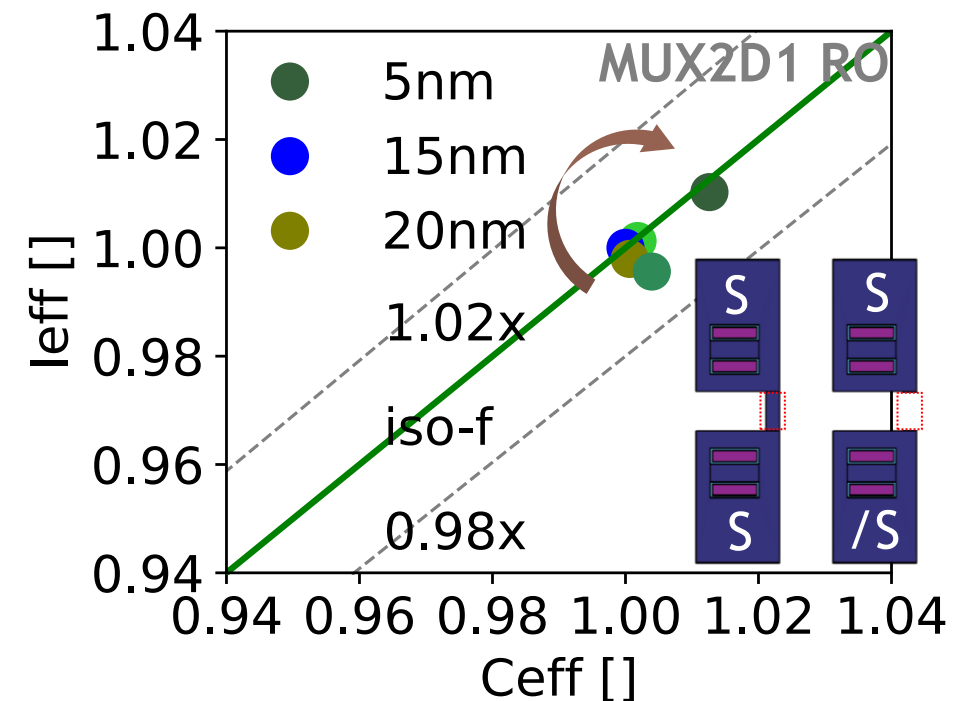
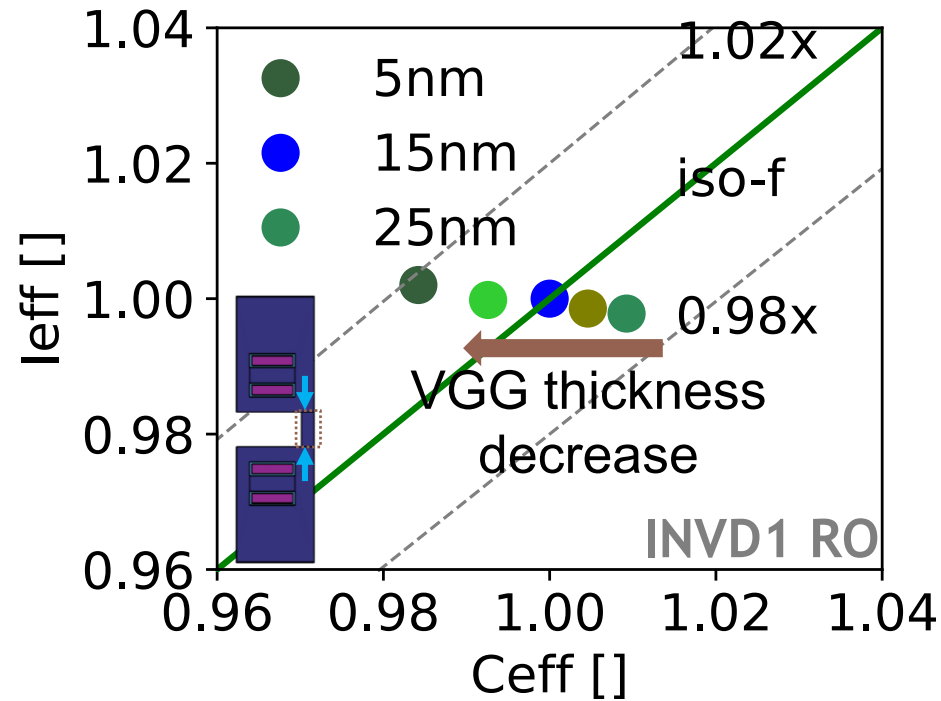
The added capacitance leads to a ~10% RO frequency reduction

Gate capacitance sensitivity evaluation



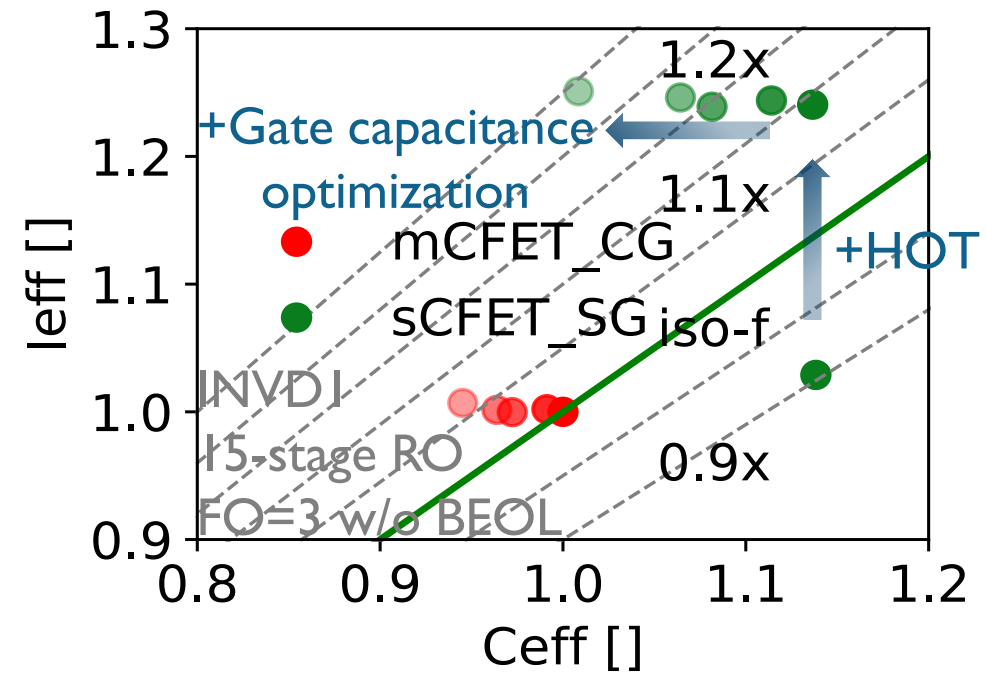
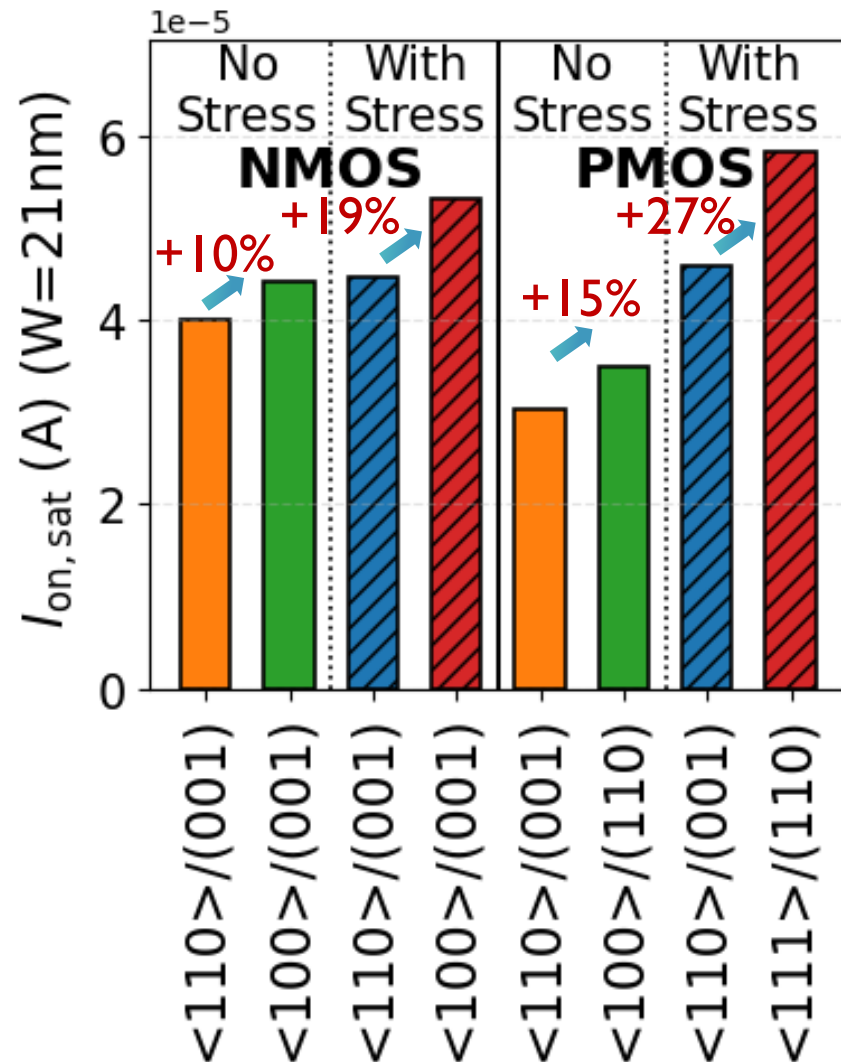
Reducing the gate, MD/BMD heights shrink the frequency gap from 10% to ~4%.

VGG thickness in Split gate



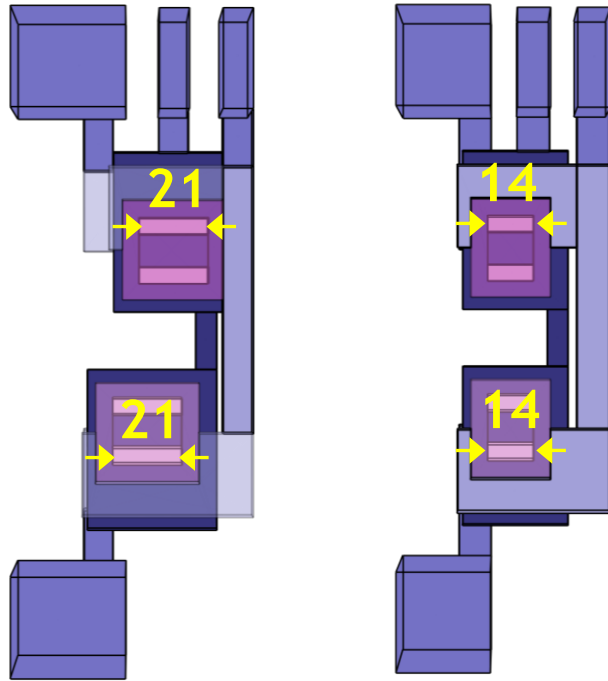
Selecting the optimal VGG thickness requires balancing connected gate and isolated gate trade-offs with process complexity

sCFET with hybrid orientation (HOT)



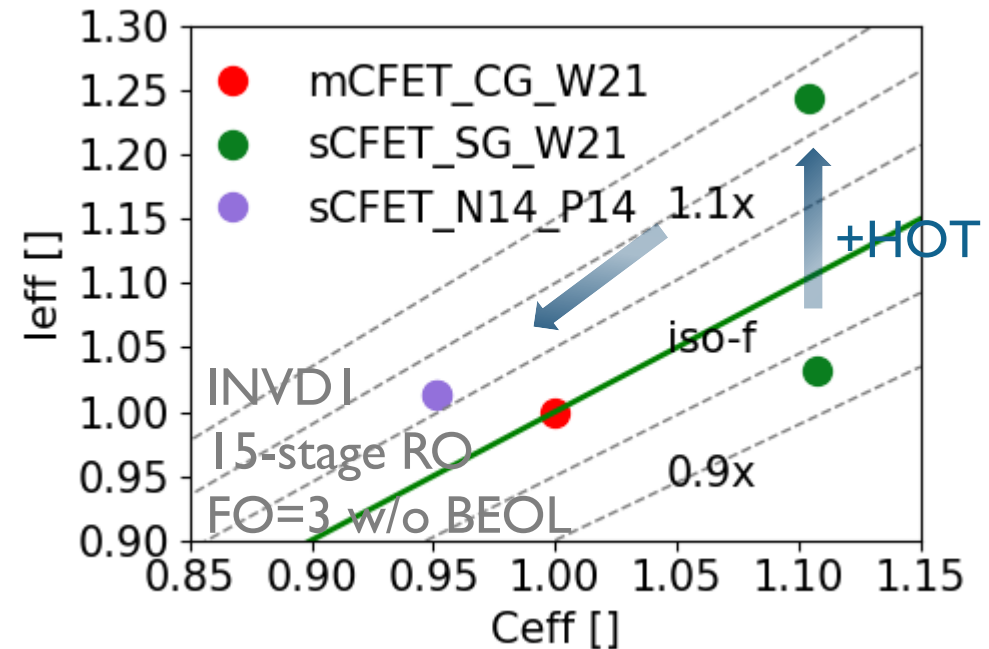
Independent NMOS/PMOS crystal-orientation optimization in sCFET delivers Ion gains shown in MC TCAD simulations.

Scaling sCFET to CH54



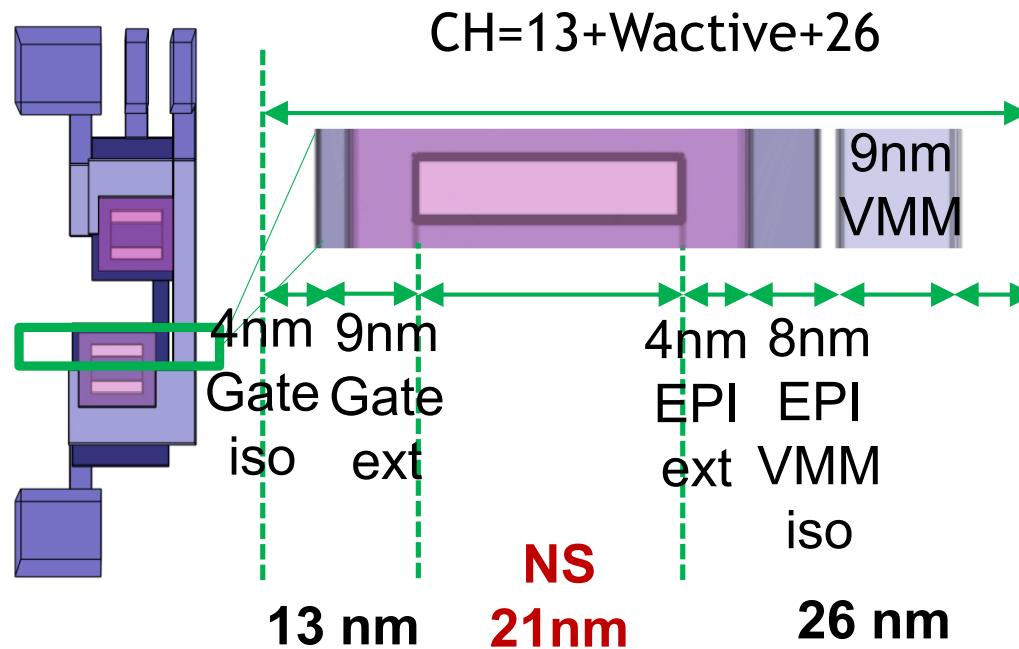
CH
60nm

CH
54nm

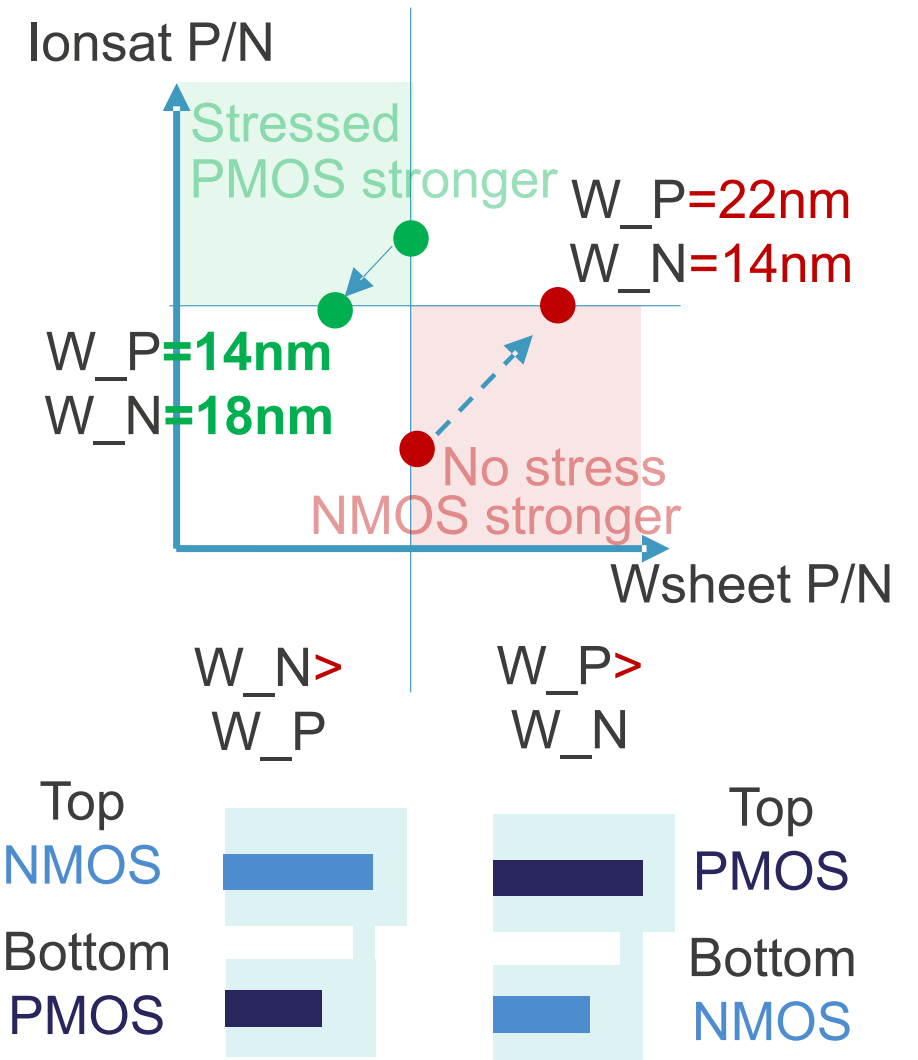


Reducing sheet width to 14nm helps scaling CH to 54nm, M0 pitch 18nm.

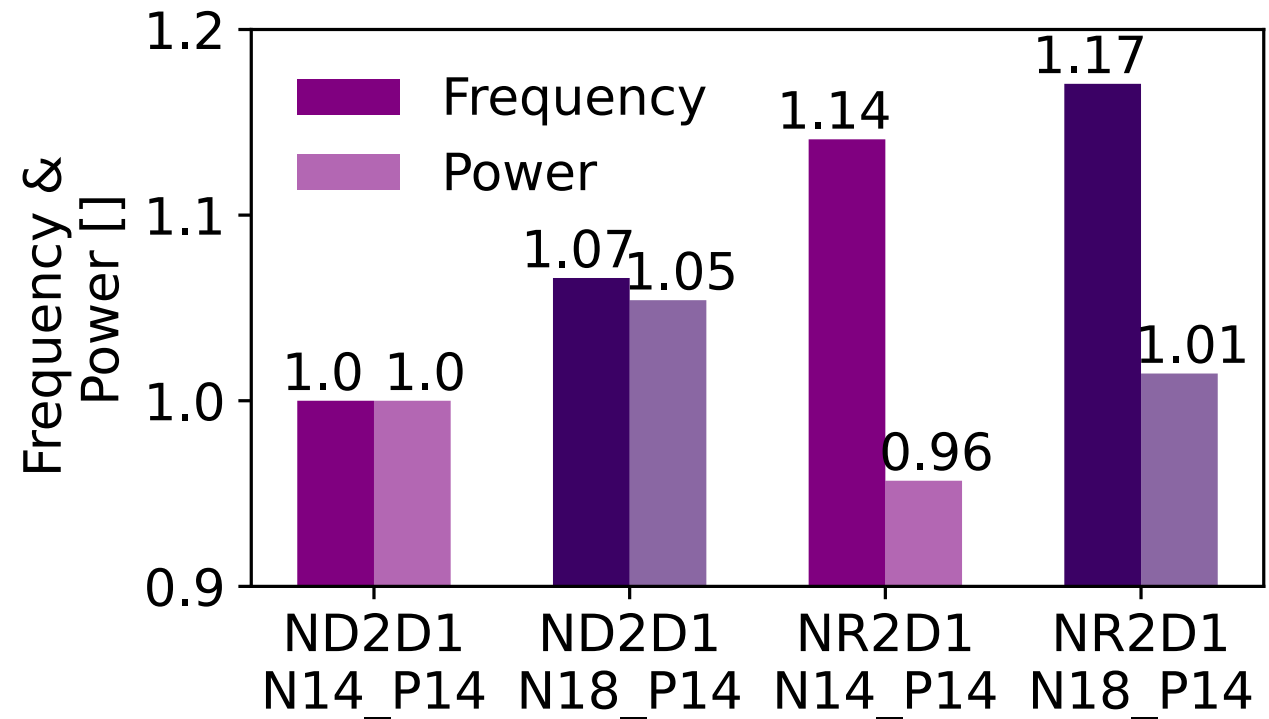
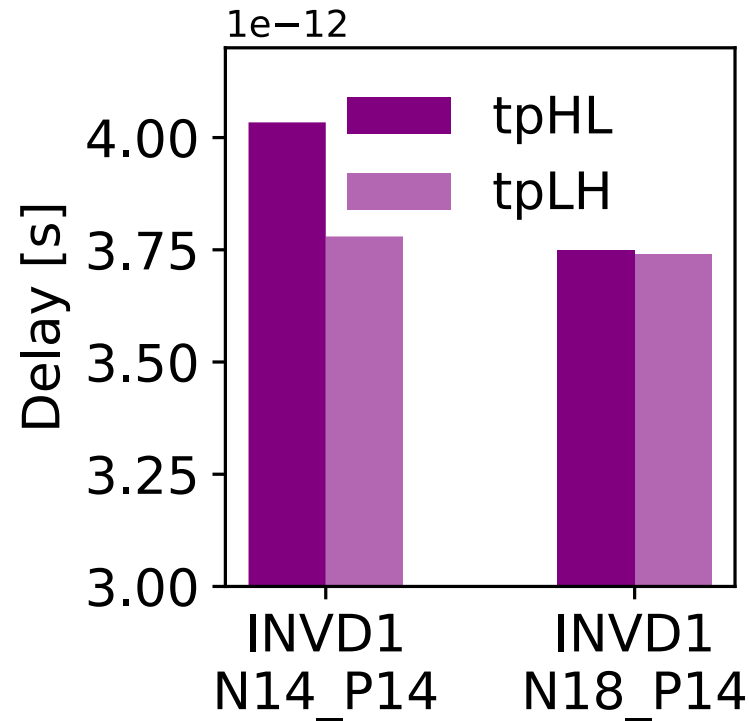
Asymmetrical sheet width



Top device can use an active sheet width that is 8 nm larger

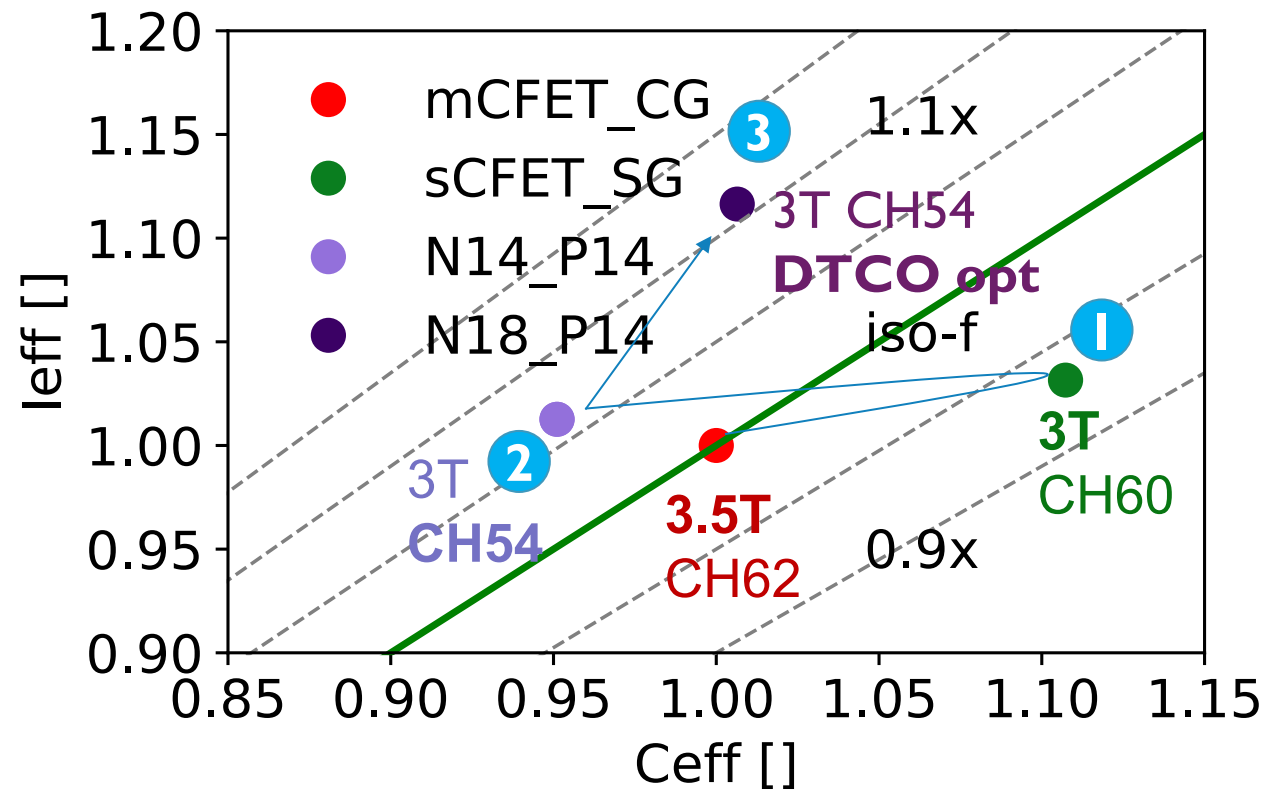


Asymmetrical sheet width



- Asymmetrical sheet width mitigates rise/fall delay imbalance
- NMOS sheet-width enlargement provides larger frequency gains (~7%) in NAND cells

DTCO optimal



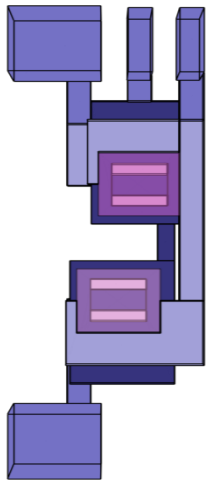
DTCO flow

- 1** +split gate,
+dense BM0
+VMM
- 2** +C opt
+HOT
+W opt
- 3** +Hybrid W opt

Conclusions

Double-flip sCFET

- split-gate + dense BM0 + VMM
 - 3T library design + HOT
 - asymmetrical sCFET
- ➔ superior PPA
 - ➔ highest library efficiency
 - ➔ scaled Cell Height to 54nm
 - ➔ hybrid sheet-width optimization



Thank you!



NanoIC

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